

Austin M. Newby

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<https://anewby0.github.io/>

Objective

Creative and driven 2nd-year Electrical Engineering student seeking expanded experiences in digital signal processing and energy production. Looking forward to applying backgrounds in teamwork, leadership, and technical skills in a professional environment.

Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Electrical Engineering, GPA 3.75

August 2025 – Present

Expected Graduation: May 2028

Skills

Programming: Java, Python, JavaScript, C#, C++, GDScript, Visual Python

Platforms: Linux (Ubuntu, Kali), Arduino

Web Development: HTML, CSS

Software: Git/GitHub, Unity, Godot, FL Studio

Experience

TJMaxx | Edmond, OK

July 2023 – August 2025, May 2026 – August 2026

Backroom Associate (20hr/week)

- Mentored 2 associates with store procedures; assisted customers by lifting heavy merchandise, providing in-store directions, and informing them of promotions and charity events.

Projects

OLED Moon Phase Display

Summer 2026

- Utilized ESP-32 DevKit w/ SSD1306 OLED to automatically fetch and display moon phase data every 12 hours using C++.
- Created bitmap representations of phases, calculated device battery-life, applied JSON parsing and Wifi security considerations.

Midi-Controlled LED Ring

Spring 2026

- Utilized Arduino UNO w/ 12-ct Neopixel ring visualizing Midi-keyboard inputs with C++ coded RGB animations.
- Employed 3rd-party tools (hairlessMidi, loopMidi, FL Studio) to route Midi data, parsed Midi CC and Note data w/ C++.

GGJ Global Game Jam 2026

Spring 2026

Programmer/Designer

GGJ is an annual 48 hr game development event. Attended in person judging and ceremonies at Georgia State University.

- Collaborated in a 4 person hybrid team (3 in-person 1-online) giving feedback on all aspects from initial design to final product.
- Learned to resolve Git conflicts with precision, and cooperative programming tactics.
- Adapted to time constraints, noting the importance of flexibility especially in stressful situations.

Various Game Design Projects | VGDev Club @GT

Fall 2025 – Present

Programmer/Composer

VGDev is a team-based student org. Each semester 6 teams pitch game ideas. Club members can choose to participate in any number of teams and are afforded tasks by the team leads. Engaged in 4 teams, utilizing the Godot and Unity game engines.

- Attended weekly 2 hour progress meetings and team socials; met 1-on-1 with team leads and created personal connections.
- Awarded a “Deleonic,” a one-time recognition of exceptional performance and participation for work on *Bird Brain Bash*.
- Designed fundamental audio, scene-loading, text, and UI systems employed by other members, enabling major advancements.

Relevant Coursework

Intro Signal Processing: Sampling Theorem, Filtering, Frequency response, Discrete Fourier Transform, Z Transform.

Digital System Design: Boolean algebra, circuits, logic gates, creating schematics, and computer logic

Physics II: Electromagnetic interactions, experimental design, and field theory

Differential Equations: Numerical/analytic solutions of differential equations, emphasis on modeling.

Engineering Software Design: Object-oriented design, numerical analysis, simulation and graphical presentation of results.

Circuit Analysis: DC and AC circuit theory and analysis

Digital Design Lab: Project/team design, CAD tools, logic synthesis, assembly language programming.

Leadership

CONNECT (First Year Leadership Organization) | Team Member

Fall 2025 – Spring 2026

- Attended weekly leadership meetings within the organization to develop cooperative and interpersonal skills
- Collaborated with a 4-person team to budget and schedule student outreach events to reduce stress during finals.

Activities

Tabletop Roleplaying Group | Gamemaster/Player

Fall 2022 – Present

- Managed, designed, and scheduled a 1+ year campaign and 4 short form games in various complex game systems.
- Cooperated with friends to develop original settings and other creative works.

Personal Music Composition and Performance

Spring 2020 – Present

- Created digital music utilizing self-taught skills in digital audio workstations; submitted 4 songs for online competitions.
- Applied knowledge gained through percussion instruction in performing instrumental solos and ensembles at state events.